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 - c. Move files/directories
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 - e. Print the current directory tree
 - f. List contents of the current directory
 - g. Sort files in the current directory by creation date
 - 3) User Interface
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1) Project Design

- *FileSystemElement*: An abstract class with common attributes and methods for both files and directories. Attributes include:
 - name: The file or directory name.
 - type: Type of the file (file or directory).
 - dateCreated: The creation timestamp using Java's Timestamp class.
 - parent: A reference to the parent directory.
- *File*: Extends from *FileSystemElement* and represents a file.
- *Directory*: Extends from *FileSystemElement* and represents a directory that can contain other files or directories. It must manage a list of *FileSystemElement* objects representing its children. It includes:
 - children: A Linked List containing objects of *FileSystemElement*.
- *FileSystem*: Manages the root directory and contains methods for user interactions such as navigating the directory tree, and managing files and directories.
- *Main*: Start point of the project.

2) Features

a. Create files/directories

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 3

Current directory: root

Create a file or directory (f/d): f
Enter a file name: 1.txt
File successfully created: 1.txt
Press enter to proceed...

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 3

Current directory: root

Create a file or directory (f/d): d
Enter a directory name: a
Directory successfully created: a/
Press enter to proceed...

b. Delete files/directories

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 4

Current directory: root

Delete a file or directory (f/d): f
Enter the file name: 1.txt
Press enter to proceed...

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 4

Current directory: root

Delete a file or directory (f/d): d
Enter the directory name: a
Press enter to proceed...

c. List contents of the current directory

```
==== File System Management Menu ====
```

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

```
-> Select an option: 2
```

```
-----  
Current directory: root
```

```
Listing contents:
```

```
-----  
* a/  
* b/  
test.txt
```

```
Press enter to proceed...
```

```
==== File System Management Menu ====
```

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

```
-> Select an option: 2
```

```
-----  
Current directory: root/a
```

```
Listing contents:
```

```
-----  
* c/  
dirA.txt
```

```
Press enter to proceed...
```

d. Change current directory

```
-----  
==== File System Management Menu ====
```

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

```
-> Select an option: 1
```

```
-----  
Current directory: root/a
```

```
Enter new directory path: c  
Press enter to proceed...
```

```
-----  
==== File System Management Menu ====
```

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

```
-> Select an option: 1
```

```
-----  
Current directory: root/a/c
```

```
Enter new directory path: ../..  
Press enter to proceed...
```

```
-> Select an option: 2
-----
Current directory: root

Listing contents:
-----
* a/
* b/
test.txt
Press enter to proceed...
-----

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 1
-----
Current directory: root

Enter new directory path: b
Press enter to proceed...
-----
```

```
-----
==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 1
-----
Current directory: root/b

Enter new directory path: root/a/c
Press enter to proceed...█
```

e. Move files/directories

```
Current directory: root/a/c
Listing contents:
-----
test.txt
Press enter to proceed...
-----
==== File System Management Menu ====
1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit
-> Select an option: 5
-----
Current directory: root/a/c
Enter the name of the file/directory: test.txt
Enter the full path of the directory: root/b
Press enter to proceed...

-> Select an option: 2
-----
Current directory: root/a/c
Listing contents:
-----
Directory is empty...
Press enter to proceed...
-----
==== File System Management Menu ====
1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit
-> Select an option: █

-> Select an option: 1
-----
Current directory: root/a/c
Enter new directory path: root/b
Press enter to proceed...
-----
==== File System Management Menu ====
1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit
-> Select an option: 2
-----
Current directory: root/b
Listing contents:
-----
test.txt
Press enter to proceed... █
```

f. Search for files/folders

```
==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 6
-----
Current directory: root/b

Enter search query: test.txt
Searching from root...
Found: root/test.txt
Press enter to proceed...
```

g. Print the current directory tree

```
-----
Current directory: root/a/c

Listing contents:
-----
* d/
dirC.txt
Press enter to proceed...
-----

==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 7
-----
Current directory: root/a/c

* root/
  * a/ (Current Directory)
    * c/ (Current Directory)
      * d/
      dirC.txt
Press enter to proceed...
```

h. Sort files in the current directory by creation date

```
==== File System Management Menu ====

1. Change directory
2. List directory
3. Create file/directory
4. Delete file/directory
5. Move file/directory
6. Search file/directory
7. Print directory tree
8. Sort contents
9. Exit

-> Select an option: 8
-----
Current directory: root/a/c

* d/ (2024-04-22 23:07:01.703)
dirC.txt (2024-04-22 23:07:13.283)
dirC_2.txt (2024-04-22 23:10:26.147)
* new_dir/ (2024-04-22 23:10:41.062)
Press enter to proceed...
```